Niranjan Kumar G S

AI Engineer | Automation Architect

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Professional Summary:

AI Automation Architect with 9+ years of experience transforming traditional QA into intelligent, self-learning systems powered by LLMs, Agentic AI, and RAG-based memory.

Architected a Cognitive Autonomous QA Platform that replaces script-based automation with a closed-loop reasoning system (ReAct) capable of planning, execution, self-healing, and continuous learning. Leveraged LLM orchestration, vector memory, and Vision AI to build failure-aware systems that adapt to dynamic applications with minimal human intervention.

Strong expertise in system design, failure handling, and scalable automation architectures, with proven impact in reducing maintenance by 80%, accelerating release cycles by 80%, and detecting 75% of defects pre-release.

Technical Skills

- AI/LLM: Agentic AI (ReAct), Prompt Engineering (Structured/JSON), RAG, LLM Orchestration, Tool Calling, Vector Embeddings
- AI Tools & Platforms: Ollama, LLaMA, Qwen, DeepSeek, Gemini (Vertex AI), OpenAI APIs, Hugging Face
- Programming: Python, Java, JavaScript, Shell, Bash
- Automation: Playwright, Selenium, Cypress, Appium, PactumJS, RestAssured
- Frameworks: TestNG, JUnit, PyTest, Mocha, Jest, Robot Framework, Cucumber (BDD), TDD
- API Testing: REST APIs, Postman, Swagger
- DevOps & CI/CD: Jenkins, GitHub Actions, Azure DevOps, TeamCity, Git, GitHub, GitLab
- Cloud & Infra: AWS, GCP, Docker, Kubernetes, Grafana
- Data & Storage: PostgreSQL, MySQL, Vector DB concepts (FAISS/Chroma)
- Tools & Platforms: JIRA, Confluence, QTest, Jama, Perforce

Work Experience

Senior Software Engineer – AI Automation (SDET II) | DP World | Feb 2022 – Present | *Dubai*

- Built and scaled an AI-first automation platform across QA pipelines, driving the transition from script-based frameworks to agentic, self-learning systems.
- Defined architecture and design decisions across LLM orchestration, memory systems, and execution pipelines to ensure scalability and reliability.
- Architected and developed a Cognitive Autonomous QA Platform using Agentic AI (ReAct), LLM orchestration, and RAG-based memory for intelligent and adaptive test automation.
- Designed a closed-loop execution engine (Plan → Think → Act → Observe → Learn) enabling autonomous decision-making and continuous learning.
- Built a self-healing automation system leveraging RAG memory, DOM analysis, and similarity-based selector recovery, significantly reducing flakiness and maintenance overhead.
- Integrated Vision AI for UI validation and automated error detection, along with a failure-aware execution pipeline incorporating retry strategies and root-cause analysis.
- Orchestrated multi-model LLM workflows (LLaMA, Qwen, DeepSeek via Ollama) and integrated with CI/CD pipelines for scalable, fully automated regression execution.

Key Impact: Reduced automation maintenance by 80%, accelerated release cycles by 80%, and enabled detection of 75% defects pre-release.

System Automation Engineer – Skylo Technologies *Aug 2020 – Feb 2022 | Bangalore*

- Built and executed tests using Python, Selenium, Appium, Playwright
- Developed dashboards using Dash + Plotly, performed DB validation on PostgreSQL
- Supported IoT modules, message queues (RabbitMQ), and internal tools
- CI/CD via Jenkins, worked across hardware and cloud

QAE Automation Engineer – Sony (via GlobalEdge) *Sep 2018 – Aug 2020 | Bangalore*

- Automated 100+ test scenarios for PlayStation consoles using PyTest and Selenium.
- Coordinated Agile QA cycles with Jira + QTest; integrated with Jenkins CI.
- Collaborated with Sony R&D to ensure 95%+ feature coverage through automation.

Python Automation Engineer – Intel (via UST Global) *Mar 2018 – Sep 2018 | Bangalore*

- Automated CLI and GUI testing for Intel motherboard tools using Python and Selenium.
- Created JSON-driven test definitions; generated real-time HTML reports.
- Improved script reusability and test coverage by 80%.

Test Engineer – SanDisk (via Randstad) *Sep 2017 – Mar 2018 | Bangalore*

- Automated firmware validation for mobile platforms: Samsung, Huawei, LG
- Automated firmware testing using Python and proprietary tools; reduced manual effort by 60%.
- Used proprietary tools and logs for test reporting and debugging

Awards

- **Talent Award – Action Oriented**, DPWorld
- **Kudos Award Winner**, DPWorld
- **Star of the Year**, Skylo Technologies
- **Lime Lite Award**, Global Edge

Certifications

- **Generative AI & Agents for Software Testing** – Udemy
- **Google IT Automation with Python** – Coursera
- **Data Structures & Algorithms in Python** – NPTEL (IIT Madras)
- Multiple Courses: Git/GitHub, OS Automation, Debugging, Cloud Basics (Coursera)

Publications & Articles

- One of the Key Challenges in UI Automation: Maintainability vs. Flakiness
- Mastering Polymorphism: A Key to Effective Object-Oriented Testing
- Mastering Software Testing Types: Smoke, Sanity & Regression
- Playwright Test Automation with TestNG & Page Object Model

Education

🎓 **B.E. – Electronics & Communication**, VTU (2012–2016)
🎓 **Diploma – E&C**, Govt. Board of Technical Education (2009–2012)